
3.3 Primitive Rules of Proof

Comment. We introduce six new primitive rules of proof: universal elimination, universal introduction, existential introduction, existential elimination, identity introduction, and identity elimination. To allow succinct statements of the first four of these, the notions of universalization, existentialization, and instance are defined.

universalization

Definition. A **UNIVERSALIZATION** of a sentence with respect to a given name occurring in the sentence is obtained by the following two steps:

- (1) Replace all occurrences of the name in the sentence by a variable α , where α does not already occur in the sentence.
- (2) Prefix $\forall\alpha$ to the open formula resulting from step 1.

Examples.

Universalizations of

$(Fa \rightarrow Ga)$

include

$\forall x(Fx \rightarrow Gx)$

and $\forall y(Fy \rightarrow Gy)$.

Universalizations of

Faa

include

$\forall xFxx$

and $\forall yFyy$.

existential- lization

Definition. An **EXISTENTIALIZATION** of a sentence with respect to a given name occurring in the sentence is obtained by the following two steps:

- (1) Replace at least one occurrence of the name in the sentence by a variable α , where α does not already occur in the sentence.
- (2) Prefix $\exists\alpha$ to the open formula resulting from step 1.

Comment. Notice the difference between step 1 in the definition of universalization and step 1 in the definition of existentialization. Universalization requires replacement of *all* occurrences of the name by with α .

Examples.

Existentializations of

$(Fa \rightarrow Ga)$

include

$\exists x(Fx \rightarrow Gx),$

$\exists x(Fa \rightarrow Gx),$

and $\exists y(Fy \rightarrow Ga).$

Existentializations of

Faa

include

$\exists xFxx,$

$\exists xFax,$

and $\exists yFya$.

instance

Definition. An **INSTANCE** of a universally or existentially quantified sentence is the result of the following two steps:

- (1) Remove the initial quantifier.
- (2) In the open formula resulting from step 1, uniformly replace all occurrences of the unbound variable by a name.

Comment. This is called **INSTANTIATING** the sentence. The name is called the **INSTANTIAL NAME**.

Examples.

The sentence

$\forall xFx$

has instances

$Fa, Fb, Fc,$ etc.

The sentence

$\exists x(Fx \ \& \ Gx)$

has instances

$(Fa \ \& \ Ga), (Fb \ \& \ Gb), (Fc \ \& \ Gc),$ etc.

The sentence

$$\exists x \forall y (Fxy \rightarrow Gy)$$

has instances

$$\forall y (Fay \rightarrow Gy), \forall y (Fby \rightarrow Gy), \text{ etc.}$$

Exercise 3.3.1* Pair wffs and their instances from the list of sentences below. Some formulas may appear in several pairs. Others may appear in none.

- | | |
|------|--|
| i | $\forall x Fax$ |
| ii | $\exists x (Fxa \ \& \ \forall y Gyxa)$ |
| iii | $\exists x Fax$ |
| iv | Fab |
| v | $\exists y \forall x Fyx$ |
| vi | $\exists zx (Fxz \ \& \ \forall y Gyxa)$ |
| vii | $\forall xy Fxy$ |
| viii | $\forall x Fxa$ |
| ix | $\exists zx (Fxz \ \& \ \forall y Gyxz)$ |
| x | $Fba \ \& \ \forall y Gyba$ |

Comment. The primitive rules of proof for predicate logic include all the primitive rules from chapter 1. There are also introduction and elimination rules for the two quantifiers and for the identity symbol. Two of the new rules have conditions that must be met for the application of the rules to be correct.

universal-elim Given a universally quantified sentence (at line m), conclude any instance of it.

Condition: None.

Annotation: **m** $\forall E$

Assumption set: same as line m .

Also known as: Universal Instantiation.

Examples.

(a)

1	(1)	$\forall xFx$	A
1	(2)	Fa	1 $\forall E$
1	(3)	Fb	1 $\forall E$

(b)

1	(1)	$\forall yRyy$	A
1	(2)	Rbb	1 $\forall E$

universal-intro Given a sentence (at line m) containing at least one occurrence of a name, conclude a universalization of the sentence with respect to that name.

Condition: The name in question must not occur in any assumptions in m 's assumption set.

Annotation: **m** $\forall I$

Assumption-set: same as line m .

Also known as: Universal Generalization.

Examples.

(a)

1	(1)	$\forall xFx$	A
1	(2)	Fb	1 $\forall E$
1	(3)	$\forall xFx$	2 $\forall I$
1	(4)	$\forall yFy$	2 $\forall I$

(b)

1	(1)	$\forall x(Fx \rightarrow Gx)$	A
1	(2)	$Fa \rightarrow Ga$	1 $\forall E$
3	(3)	$\forall xFx$	A
3	(4)	Fa	3 $\forall E$
1,3	(5)	Ga	2,4 $\rightarrow E$
1,3	(6)	$\forall xGx$	5 $\forall I$

Example of violation of the $\forall I$ condition.

(c)

1	(1)	$\forall x(Fx \rightarrow Gx)$	A
2	(2)	Fa	A
1	(3)	$Fa \rightarrow Ga$	1 $\forall E$
1,2	(4)	Ga	2,3 $\rightarrow E$
1,2	(5)	$\forall xGx$	4 $\forall I$

wrong!

Comment. Ordinarily we cannot conclude $\forall xFx$ merely from Fa —the fact that one thing is F doesn't guarantee that everything is F ! The condition on $\forall I$ ensures that we do not make this mistake. If the sentence Fa is true, and furthermore *would still be true no matter what the name denotes*, then clearly everything is F , so we can conclude $\forall xFx$. When the condition on $\forall I$ is met, then we are in such a situation: if we

prove Fa from assumptions that do not contain the name a and hence say nothing in particular about its referent, then we could just as well have used a different name, say b , and proved Fb . In fact, when the condition on $\forall I$ is met, any proof of Fa can be turned into a proof of Fb just by replacing any involved occurrences of the name a by the name b . This is sufficient to guarantee that everything is F ; hence, we can conclude $\forall xFx$.

existential- intro

Given a sentence (at line m) containing at least one occurrence of a name, conclude an existentialization of that sentence with respect to that name.

Condition: None.
Annotation: **$m \exists I$**
Assumption-set: same as line m .
Also known as: Existential Generalization.

Examples.

(a)

1	(1)	Fa	A
1	(2)	$\exists xFx$	1 $\exists I$

(b)

1	(1)	$\forall x(Fx \rightarrow Gx)$	A
2	(2)	Fa	A
1	(3)	$Fa \rightarrow Ga$	1 $\forall E$
1,2	(4)	Ga	2,3 $\rightarrow E$
1,2	(5)	$Fa \& Ga$	2,4 $\& I$
1,2	(6)	$\exists x(Fx \& Gx)$	5 $\exists I$

(c)

1	(1)	$\forall x Fx$	A
1	(2)	$\exists y \forall x Fyx$	1 $\exists I$

existential- elim

Given a sentence (at line m) and an assumption (at line i) that is an instance of some existentially quantified sentence that is present (at line k), conclude the given sentence again.

Condition: The instantial name at line i must not appear in the sentence at line k or in the sentence at line m . Also, it must not appear in any of the assumptions belonging to the assumption set at line m , other than the instance i itself.

Annotation: $k, m \exists E (i)$

Assumption set: all assumptions at line m other than i , and all assumptions at line k .

Examples.

(a)

1	(1)	$\exists x Fx$	A
2	(2)	Fa	A
2	(3)	$Fa \vee Ga$	2 $\vee I$
2	(4)	$\exists x (Fx \vee Gx)$	3 $\exists I$
1	(5)	$\exists x (Fx \vee Gx)$	1,4 $\exists E (2)$

(b)

1	(1)	$\exists x(Fxx \rightarrow P)$	A
2	(2)	$Faa \rightarrow P$	A
3	(3)	$\forall xFxx$	A
3	(4)	Faa	3 $\forall E$
2,3	(5)	P	2,4 $\rightarrow E$
1,3	(6)	P	1,5 $\exists E$ (2)

Examples of violation of $\exists E$ condition.

(a)

	1	(1)	$\exists xFx$	A
	2	(2)	Fa	A
	3	(3)	Ga	A
	2,3	(4)	$Fa \& Ga$	2,3 $\&I$
	2,3	(5)	$\exists x(Fx \& Gx)$	4 $\exists I$
wrong!	1,3	(6)	$\exists x(Fx \& Gx)$	1,5 $\exists E$ (2)

(b)

	1	(1)	$\exists xFx$	A
	2	(2)	Fa	A
wrong!	1	(3)	Fa	1,2 $\exists E$ (2)

(c)

	1	(1)	$\exists xFax$	A
	2	(2)	Faa	A
	2	(3)	$\exists xFxx$	2 $\exists I$
wrong!	1	(4)	$\exists xFxx$	1,3 $\exists E$ (2)

Comment. If all we know is that *something* is F, we are not entitled to reason as if we know *what* it is that is F. As in the case of $\forall I$, a use of $\exists E$ that meets the conditions above and uses a certain instantial name can be turned into a proof of the same conclusion from the same assumptions but using any different instantial name. This shows that the conclusion does not rest on any assumptions about the actual identity of the thing that is said to exist. That is, if we apply $\exists E$ to $\exists xFx$ by discharging the assumed instance Fa , the conditions ensure that we do not mistakenly use any information about the referent of ‘a’ *in particular*. After all, $\exists xFx$ says only that something is F—it doesn’t tell us *which* individual is F.

identity-intro Conclude any sentence of the form $\alpha=\alpha$.

Condition: None.
Annotation: =I
Assumption set: Empty.

Example.

(1) $c=c$ =I

Comment. An identity statement of the form $\alpha=\alpha$, like a theorem, requires no assumptions to justify its assertion.

identity-elim Given a sentence ϕ (at line m) containing a name α , and another sentence (at line n) that is an identity statement containing α and another name β , conclude a sentence that is the result of replacing at least one occurrence of α in ϕ with β .

Condition: None.

Annotation: **$m, n = E$**

Assumption set: The union of the assumption sets at lines m and n .

Also known as: Leibniz's Law, Substitutivity of Identity

Examples.

(a)

1	(1)	Fa	A
2	(2)	a=b	A
1,2	(3)	Fb	1,2 =E

(b)

1	(1)	Fa & Ga	A
2	(2)	b=a	A
1,2	(3)	Fb & Ga	1,2 =E
1,2	(4)	Fb & Gb	1,2 =E

(c)

1	(1)	$\forall x(Fxa \rightarrow x=a)$	A
2	(2)	Fba	A
1	(3)	$Fba \rightarrow b=a$	1 $\forall E$
1,2	(4)	b=a	2,3 $\rightarrow E$
1,2	(5)	$\forall x(Fxb \rightarrow x=b)$	1,4 =E

Comment. The rule of identity elimination is not regarded as valid in all contexts. For instance, if Frank believes that Mark Twain is a novelist then, even though Twain=Clemens, it does not follow that he believes Samuel Langhorne Clemens is a novelist (if, for example, he has heard the name "Twain" but never "Clemens"). For historical reasons, contexts where the rule fails, such as belief reports, are called *intensional* contexts in contrast to the *extensional* contexts provided by the ordinary predicates which the language developed in this chapter is intended to represent.

Exercise 3.3.2 Prove the following sequents, using the primitive rules of predicate logic. You may also use derived sentential rules.

- S87* $\exists x(Gx \ \& \ \sim Fx), \forall x(Gx \rightarrow Hx) \vdash \exists x(Hx \ \& \ \sim Fx)$
 S88* $\exists x(Gx \ \& \ Fx), \forall x(Fx \rightarrow \sim Hx) \vdash \exists x \sim Hx$
 S89* $\forall x(Gx \rightarrow \sim Fx), \forall x(\sim Fx \rightarrow \sim Hx) \vdash \forall x(Gx \rightarrow \sim Hx)$
 S90 $\exists x(Fx \ \& \ Ga), \forall x(Fx \rightarrow Hx) \vdash Ga \ \& \ \exists x(Fx \ \& \ Hx)$
 S91* $\forall x(Gx \rightarrow \exists y(Fy \ \& \ Hy)) \vdash \forall x \sim Fx \rightarrow \sim \exists z Gz$
 S92* $\forall x(Gx \rightarrow Hx \ \& \ Jx), \forall x(Fx \vee \sim Jx \rightarrow Gx) \vdash \forall x(Fx \rightarrow Hx)$
 S93* $\forall x(Gx \ \& \ Kx \leftrightarrow Hx), \sim \exists x(Fx \ \& \ Gx) \vdash \forall x \sim (Fx \ \& \ Hx)$
 S94 $\forall x(Gx \rightarrow Hx), \exists x((Fx \ \& \ Gx) \ \& \ Mx) \vdash \exists x(Fx \ \& \ (Hx \ \& \ Mx))$
 S95* $\forall x(\sim Gx \vee \sim Hx), \forall x((Jx \rightarrow Fx) \rightarrow Hx) \vdash \sim \exists x(Fx \ \& \ Gx)$
 S96* $\sim \exists x(\sim Gx \ \& \ Hx), \forall x(Fx \rightarrow \sim Hx) \vdash \forall x(Fx \vee \sim Gx \rightarrow \sim Hx)$
 S97 $\forall x \sim (Gx \ \& \ Hx), \exists x(Fx \ \& \ Gx) \vdash \exists x(Fx \ \& \ \sim Hx)$
 S98 $\exists x(Fx \ \& \ \sim Hx), \sim \exists x(Fx \ \& \ \sim Gx) \vdash \sim \forall x(Gx \rightarrow Hx)$
 S99 $\forall x(Hx \rightarrow Hx \ \& \ Gx), \exists x(\sim Gx \ \& \ Fx) \vdash \exists x(Fx \ \& \ \sim Hx)$
 S100 $\forall x(Hx \rightarrow \sim Gx), \sim \exists x(Fx \ \& \ \sim Gx) \vdash \forall x \sim (Fx \ \& \ Hx)$
 S101* $\forall x(Fx \leftrightarrow Gx) \vdash \forall x Fx \leftrightarrow \forall x Gx$
 S102* $\exists x Fx \rightarrow \forall y(Gy \rightarrow Hy), \exists x Jx \rightarrow \exists x Gx \vdash \exists x(Fx \ \& \ Jx) \rightarrow \exists z Hz$
 S103 $\exists x Fx \vee \exists x Gx, \forall x(Fx \rightarrow Gx) \vdash \exists x Gx$

- S104 $\forall x(Fx \rightarrow \sim Gx) \vdash \sim \exists x(Fx \& Gx)$
- S105* $\forall x(Fx \vee Hx \rightarrow Gx \& Kx), \sim \forall x(Kx \& Gx) \vdash \exists x \sim Hx$
- S106 $\forall x(Fx \& Gx \rightarrow Hx), Ga \& \forall x Fx \vdash Fa \& Ha$
- S107* $\forall x(Fx \leftrightarrow \forall y Gy) \vdash \forall x Fx \vee \forall x \sim Fx$
- S107 $\forall y(Fa \rightarrow (\exists x Gx \rightarrow Gy)), \forall x(Gx \rightarrow Hx), \forall x(\sim Jx \rightarrow \sim Hx)$
 $\vdash \exists x \sim Jx \rightarrow \sim Fa \vee \forall x \sim Gx$
- S109* $\forall x(Dx \rightarrow Fx) \vdash \forall z(Dz \rightarrow (\forall y(Fy \rightarrow Gy) \rightarrow Gz))$
- S110 $\exists x Fx \leftrightarrow \forall y(Fy \vee Gy \rightarrow Hy), \exists x Hx, \sim \forall z \sim Fz \vdash \exists x(Fx \& Hx)$
- S111* $\forall x Fx \vdash \sim \exists x Gx \leftrightarrow \sim(\exists x(Fx \& Gx) \& \forall y(Gy \rightarrow Fy))$
- S112* $\forall x(\exists y Fyx \rightarrow \forall z Fxz) \vdash \forall yx(Fyx \rightarrow Fxy)$
- S113 $\exists x(Fx \& \forall y Gxy), \forall xy(Gxy \rightarrow Gyx) \vdash \exists x(Fx \& \forall y Gyx)$
- S114 $\exists x \sim \forall y(Gxy \rightarrow Gyx) \vdash \exists x \exists y(Gxy \& \sim Gyx)$
- S115 $\forall x(Gx \rightarrow \forall y(Fy \rightarrow Hxy)), \exists x(Fx \& \forall z \sim Hxz) \vdash \sim \forall x Gx$
- S116 $\forall xy(Fxy \rightarrow Gxy) \vdash \forall x(Fxx \rightarrow \exists y(Gxy \& Fyx))$
- S117* $\forall xy(Fxy \rightarrow \sim Fyx) \vdash \sim \exists x Fxx$
- S118 $\forall x \exists y(Fxy \& \sim Fyx) \vdash \exists x \sim \forall y Fxy$
- S119 $\forall y(\exists x \sim Fxy \rightarrow \sim Fyy) \vdash \forall x(Fxx \rightarrow \forall y Fyx)$
- S120 $\exists x Fxx \rightarrow \forall xy Fxy \vdash \forall x(Fxx \rightarrow \forall y Fxy)$
- S121 $a=b \vdash b=a$
- S122 $a=b \& b=c \vdash a=c$
- S123* $a=b, b \neq c \vdash a \neq c$
- S124 $Fa \& \forall x(Fx \rightarrow x=a), \exists x(Fx \& Gx) \vdash Ga$
- S125* $\forall x x=x \rightarrow \exists x Fx, \forall x(\sim Fx \vee Gx) \vdash \exists x(Fx \& Gx)$
- S126 $\forall x(Fx \rightarrow Gx), \forall x(Gx \rightarrow Hx), Fa \& \sim Hb \vdash a \neq b$
- S127* $\exists x((Fx \& \forall y(Fy \rightarrow y=x)) \& Gx), \sim Ga \vdash \sim Fa$
- S128 $\exists x \forall y((\sim Fxy \rightarrow x=y) \& Gx) \vdash \forall x(\sim Gx \rightarrow \exists y(y \neq x \& Fyx))$
- S129 $\exists x(Px \& (\forall y(Py \rightarrow y=x) \& Qx)), \exists x \sim (\sim Px \vee \sim Fx)$
 $\vdash \exists x(Fx \& Qx)$
- S130* $\forall x \exists y Gyx, \forall xy(Gxy \rightarrow \sim Gyx) \vdash \sim \exists y \forall x (x \neq y \rightarrow Gyx)$